THE CHOW RANK OF THE SIX-BY-SIX PERMANENT

Abstract. Let perm_6 be the permanent of a generic 6 by 6 matrix. Over an algebraically closed field of characteristic zero, we prove that its ordinary Chow rank is 32. The upper bound is Glynn's identity. The lower bound combines a symmetric image-span inequality with a half-defect estimate for arbitrary quotient derivative symbols of a sextic Chow term. Exact differentiation distinguishes formal subproduct spaces from the actual derivative spaces in two dependent-factor profiles. The proof computes the actual spaces and uses sufficient local rows. The finite parts have an exact rational verification. No border-rank or general-n equality is claimed.

1. Statement and permanent derivatives

Let $\mathbb{K}$ be algebraically closed of characteristic zero and let

$$V = \mathrm{span}_{\mathbb{K}}\{x_{rc} : 1 \le r, c \le 6\}, \qquad \dim V = 36.$$

For a homogeneous polynomial f, let $\mathcal{D}_j(f) \subset \mathrm{Sym}^j V$ be the span of all derivatives whose remaining degree is j, and put $E_j = \mathcal{D}_j(\mathrm{perm}_6)$.

Definition 1.1. A sextic Chow term is a nonzero product $T = \ell_1 \cdots \ell_6$. The ordinary Chow rank $\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(f)$ is the least number of such terms whose sum is f.

Theorem 1.2. *Over* $\mathbb{K}$*,*

$$\boxed{\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(\mathrm{perm}_6) = 32.}$$

We first record the permanent derivative facts used below.

Lemma 1.3 (Permanent derivative tower)**.** *One has*

$$\dim E_3 = 400, \qquad \dim E_2 = 225, \qquad \partial E_2 = V,$$

and

$$E_2^{(1)} := \{g \in \mathrm{Sym}^3 V : \partial g \subset E_2\} = E_3. \tag{1.1}$$

Moreover, every nonzero element of E_2 has quadratic matrix rank at least four, and every nonzero element of E_3 has at least nine essential variables.

Proof. The 3×3 and 2×2 subpermanents are bases of E_3 and E_2: distinct row-column blocks have disjoint matching supports. Their numbers are $\binom{6}{3}^2 = 400$ and $\binom{6}{2}^2 = 225$. Differentiating a rectangle permanent at one corner gives the opposite variable, proving $\partial E_2 = V$.

The inclusion $E_3 \subset E_2^{(1)}$ is immediate. Conversely, suppose all first derivatives of a cubic g lie in E_2. Row and column multiaffinity forces every monomial of g to be a three-edge matching. Inside a fixed 3×3 block, the rectangle relations equate coefficients of matching monomials related by a transposition. Transpositions connect all six matchings, so each block is a scalar subpermanent. This proves (1.1).

The row-column diagonal torus has distinct characters on each subpermanent basis. A generic one-parameter subgroup degenerates any nonzero element to a single basis element, while matrix and first-catalectic ranks cannot increase under specialization. A rectangle permanent has quadratic rank four. The nine first derivatives of a 3×3 permanent are independent. The two rank floors follow. ∎

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2. Small permanent-quadratic intersections

Lemma 2.1. *If $L \subset V$ has dimension at most six, then*

$$\dim(E_2 \cap \operatorname{Sym}^2 L) \leq 3.$$

If $\dim L \leq 5$, the upper bound improves to one.

Proof. For fixed a, q, the locus in $\operatorname{Gr}(a, V)$ on which $\dim(E_2 \cap \operatorname{Sym}^2 L) \geq q$ is closed, projective, and stable under the connected row-column torus. If nonempty, the Borel fixed-point theorem gives a torus-fixed point. Since the variable weight spaces are one-dimensional, such a point is a coordinate a-plane, equivalently a bipartite graph with a edges.

At a coordinate point, the intersection has one independent rectangle permanent for each four-cycle. Two distinct four-cycles have a union of at least six edges. If the union has exactly six edges, it is $K_{2,3}$ or $K_{3,2}$, which has exactly three four-cycles. Thus five edges support at most one four-cycle and six edges support at most three. □

3. Local linear algebra

Polarization gives an injective map

$$\delta : \operatorname{Sym}^3 L \longrightarrow L \otimes \operatorname{Sym}^2 L.$$

Lemma 3.1 (Kernel-preimage bound)**.** *Let $P : L \twoheadrightarrow D$ have rank d, let $S = \ker P$, and let $R \subset \operatorname{Sym}^2 L$. Then*

$$\ker((P \otimes 1)\delta) = \operatorname{Sym}^3 S,$$

and the inverse image of $D \otimes R$ has dimension at most

$$\boxed{\binom{\dim L - d + 2}{3} + d \dim R.} \tag{3.1}$$

Proof. The first equality follows in coordinates: all derivatives in directions modulo S vanish precisely when the cubic depends only on S. The inverse image of a subspace has dimension at most the kernel dimension plus the subspace dimension, which gives (3.1). □

Lemma 3.2 (Squarefree projected-symbol table)**.** *Let $\widehat{L} = \mathbb{K}^6$ with basis $z_1, \ldots, z_6$, and let $\widehat{F}, \widehat{U}$ be the squarefree quadratic and cubic spaces. Set*

$$\delta(z_a z_b z_c) = z_a \otimes z_b z_c + z_b \otimes z_a z_c + z_c \otimes z_a z_b.$$

If $P : \widehat{L} \twoheadrightarrow D$ has rank d and $R \subset \widehat{F}$ has dimension at most $r \leq 3$, then the following table is a rowwise lower bound for $\operatorname{rank}((P \otimes q_R)\delta)$*:*

(3.2)

$r \backslash d$	0	1	2	3	4	5	6
0	0	10	16	19	20	20	20
1	0	9	14	16	16	20	20
2	0	8	12	13	16	19	20
3	0	7	10	10	15	17	19.

Proof. The diagonal torus preserves δ. The orbit closure of $(\ker P, R)$ in the relevant product of Grassmannians contains a fixed point, and rank can only drop in specialization. Thus $\ker P$ may be taken as a vertex set B, and R as the span of an edge set A with at most r edges. Put $O = [6] \setminus B$.

Distinct source cubics have disjoint output supports. The number of killed cubics is

$$\binom{|B|}{3} + |O| e_A(B) + \sum_{b \in B} \binom{\deg_A(b, O)}{2} + \#\{\text{triangles of } A[O]\}. \tag{3.3}$$

Inspection of edge sets of size at most three gives (3.2). The exact replay independently checks all 45,696 coordinate pairs and records a minimizer for each entry. □

4. The half-defect estimate

Let $T = \ell_1 \cdots \ell_6 \neq 0$, and set

$$L = \operatorname{span}\{\ell_1, \dots, \ell_6\}, \quad U = \mathcal{D}_3(T), \quad u = \dim U,$$
$$F = \mathcal{D}_2(T), \quad R = F \cap E_2.$$

For a quotient $P : L \twoheadrightarrow D$ of rank d, define

$$\beta_{P,R} : U \xrightarrow{\delta} L \otimes F \xrightarrow{P \otimes q_R} D \otimes (F/R).$$

Proposition 4.1 (Half-defect quotient-symbol estimate)**.** *For every nonzero sextic Chow term and every quotient of its actual factor span,*

$$\boxed{\operatorname{rank} \beta_{P,R} + \frac{20-u}{2} \geq \frac{10}{3} d.} \tag{4.1}$$

Proof. Write $\ell = \dim L$.
Factor span at most four. Choose ℓ independent factors as coordinates. A generic diagonal one-parameter subgroup degenerates the remaining factors to unique coordinate initials, so T specializes to

$$x_1^{m_1} \cdots x_\ell^{m_\ell}, \qquad m_j \geq 1, \qquad \sum_j m_j = 6.$$

Middle rank cannot increase in the limit. Counting degree-three divisors over the positive partitions of six gives

$$\ell = 1, 2, 3, 4 \quad \Longrightarrow \quad u \geq 1, 2, 4, 8. \tag{4.2}$$

Lemma 2.1 gives $\dim R \leq 1$. For $0 < d < \ell$, Lemma 3.1 gives

$$\operatorname{rank} \beta_{P,R} \geq \max\left\{0, u - \binom{\ell - d + 2}{3} - d\right\}. \tag{4.3}$$

At $d = \ell$, a kernel cubic lies in $E_2^{(1)} = E_3$, but it is supported on at most six essential variables, contrary to Lemma 1.3. Thus the full symbol is injective. Minimizing over the conservative interval from (4.2) to $\min\{20, \binom{\ell+2}{3}\}$ gives

(4.4)

ℓ	$\operatorname{rank} \beta_{P,R} + (20-u)/2$ for $d = 0, \dots, \ell$
4	$(0, 9/2, 8, 10, 14)$
3	$(5, 15/2, 9, 12)$
2	$(8, 9, 11)$
1	$(19/2, 21/2)$.

Six independent factors. Here $u = 20$, the actual derivative spaces are the squarefree spaces, and $\dim R \leq 3$. The last row of (3.2), improved at $d = 6$ by full-symbol injectivity, gives

$$(0, 7, 10, 10, 15, 17, 20). \tag{4.5}$$

Five-dimensional factor span. The unique relation among the six factors puts T, after changing and rescaling coordinates, into exactly one form

$$T_s = x_1 x_2 x_3 x_4 x_5 (x_1 + \cdots + x_s), \qquad 1 \leq s \leq 5. \tag{4.6}$$

Lemma 2.1 gives $\dim R \leq 1$. Exact differentiation of the actual polynomial gives

(4.7)

s	1	2	3	4	5
$\dim F$	11	11	13	14	15
u	14	14	18	20	20.

The replay constructs the order-four and order-three derivative matrices of (4.6) in the ordinary monomial bases and row-reduces them over $\mathbb{Q}$.

For $s = 3$, the actual space U contains the nine squarefree cubics other than $x_1x_2x_3$. For $s = 4, 5$, it contains all ten squarefree cubics. These subspaces are stable under the diagonal torus. A nonzero rank-one quotient specializes to a coordinate functional, whose rank is the number of displayed triples through that coordinate: at least five for $s = 3$ and six for $s = 4, 5$. Quotienting by R loses at most one. Together with (3.1) and full-symbol injectivity this gives

$$\begin{array}{c|c|c} s & u & \operatorname{rank}\beta_{P,R} + (20-u)/2 \\ \hline 4,5 & 20 & (0,5,8,13,15,20) \\ 3 & 18 & (1,5,7,12,14,19). \end{array} \tag{4.8}$$

For $s = 1$, the actual U is spanned by the ten squarefree cubics and $x_1^2x_j$, $2 \le j \le 5$. If a direction $\lambda = \sum a_j\partial_{x_j}$ has $a_j \ne 0$ for some $j \ge 2$, the six squarefree cubics divisible by x_j and $x_1^2x_j$, projected modulo x_jL, give seven independent quadrics. The pure x_1 direction has rank ten.

For $s = 2$, a basis consists of the seven squarefree cubics containing at most one of x_1, x_2, the six cubics

$$x_2^2x_j + 2x_1x_2x_j, \qquad x_1x_2x_j + \tfrac{1}{2}x_1^2x_j \quad (3 \le j \le 5),$$

and $x_1x_2(x_1 + x_2)$. If $a_j \ne 0$ for some $j \ge 3$, five squarefree inputs and the two special inputs give, modulo x_jL, five distinct pair monomials and the independent quadrics $x_2^2 + 2x_1x_2$ and $x_1x_2 + \frac{1}{2}x_1^2$. If $\lambda = a_1\partial_{x_1} + a_2\partial_{x_2}$, the image contains the three pair monomials on x_3, x_4, x_5, a nonzero direction in each $\operatorname{span}\{x_1x_j, x_2x_j\}$, and the nonzero quadratic

$$a_2x_1^2 + 2(a_1 + a_2)x_1x_2 + a_1x_2^2.$$

Thus every nonzero directional map has rank at least seven, and every positive-rank quotient symbol has rank at least six after reducing by R.

For $d = 4$, let $S = \ker P$ be a line and let v lie in the symbol kernel. Its four complementary derivatives span a subspace of R. If that span were nonzero, the essential-variable space of v would have dimension at most two and would support a nonzero element of E_2, contrary to the rank-four floor in Lemma 1.3. Hence $v \in \operatorname{Sym}^3 S$, so the kernel has dimension at most one. The full symbol at $d = 5$ is injective. With (3.1), the $s = 1, 2$ row is

$$(3, 9, 9, 10, 16, 17). \tag{4.9}$$

Every entry of (4.4), (4.5), (4.8), and (4.9) is at least $10d/3$. This proves (4.1). ☐

Remark 4.2 (Actual derivative spaces)**.** The formal pair- and triple-product spans for T_2 have dimensions 12 and 17, while the actual derivative spaces have dimensions 11 and 14. For T_3, the formal triple-product span has dimension 19 while the actual middle derivative space has dimension 18. The proof above deliberately uses the formal squarefree table only in the six-independent-factor case. The unequal dimensions are retained as regression tests.

5. A symmetric image-span inequality

Lemma 5.1. *Let $A_i : W^* \to W$ be symmetric maps, let $r_i = \operatorname{rank} A_i$, and put $D = \dim \sum_i \operatorname{im} A_i$. Then*

$$\operatorname{rank}\Big(\sum_i A_i\Big) \ge 2D - \sum_i r_i. \tag{5.1}$$

Proof. Choose $B_i : \mathbb{K}^{r_i} \to W$ onto $\operatorname{im} A_i$. Symmetry gives $\ker A_i = (\operatorname{im} A_i)^\perp$, so $A_i = B_iJ_iB_i^*$ for an invertible symmetric J_i. With $B = [B_1 \ \cdots \ B_N]$ and block-diagonal J, Sylvester's inequality gives

$$\operatorname{rank}(BJB^*) \ge \operatorname{rank}(BJ) + \operatorname{rank}(B^*) - \sum_i r_i = 2D - \sum_i r_i.$$

☐

6. The global lower bound

Suppose

$$\mathrm{perm}_6 = \sum_{i=1}^{N} T_i$$

with no zero summands. Let

$$A_i = C_{3,3}(T_i), \quad U_i = \operatorname{im} A_i, \quad u_i = \dim U_i, \quad \delta_i = 20 - u_i, \quad \Delta = \sum_i \delta_i.$$

Put $U = \sum_i U_i$ and $\dim U = 400 + h$. The sum of the symmetric maps A_i is the permanent middle catalecticant, with rank 400 and image E_3. Lemma 5.1 gives

$$\boxed{h \le 10N - 200 - \frac{\Delta}{2}.} \tag{6.1}$$

Let $F_i = \mathcal{D}_2(T_i)$, $F = \sum_i F_i$, and $Q = F/E_2$. Linearity of the derivative spaces gives $E_2 \subset F$. Blockwise differentiation modulo E_2 defines

$$\widetilde{\beta} : \bigoplus_i U_i \longrightarrow V \otimes Q.$$

Its kernel consists precisely of tuples for which every first derivative of $\sum_i g_i$ lies in E_2, equivalently $\sum_i g_i \in E_3$ by Lemma 1.3. Since $E_3 \subset U$,

$$\operatorname{rank} \widetilde{\beta} = \dim(U/E_3) = h. \tag{6.2}$$

Let L_i be the actual factor span of T_i. Since $E_2 \subset F$, $\partial E_2 = V$, and $\partial F_i \subset L_i$, the spaces L_i span all 36 variables. Choose any ordering and put

$$W_i = \sum_{j \le i} L_j, \qquad d_i = \dim(W_i/W_{i-1}), \qquad \sum_i d_i = 36.$$

Let $Z_i \subset L_i \otimes Q$ be the image of the i-th block and let $H_i = \sum_{j \le i} Z_j$. Projection

$$\pi_i : W_i \otimes Q \longrightarrow (W_i/W_{i-1}) \otimes Q$$

kills H_{i-1}, hence

$$\dim H_i - \dim H_{i-1} \ge \dim \pi_i(H_i) = \dim \pi_i(Z_i). \tag{6.3}$$

On the current block, the first-factor map is

$$P_i : L_i \twoheadrightarrow L_i/(L_i \cap W_{i-1}), \qquad \operatorname{rank} P_i = d_i.$$

The natural map

$$F_i/(F_i \cap E_2) \longrightarrow F/E_2$$

is injective. Therefore $\dim \pi_i(Z_i)$ is exactly the local symbol rank in Proposition 4.1. Summing (6.3) gives

$$\boxed{h \ge \sum_i \left(\frac{10}{3} d_i - \frac{\delta_i}{2} \right) = 120 - \frac{\Delta}{2}.} \tag{6.4}$$

Comparing (6.1) and (6.4) cancels the complete individual middle-rank defect and yields $N \ge 32$.

For the reverse inequality, Glynn's identity [1] gives

$$\mathrm{perm}_6 = \frac{1}{32} \sum_{\epsilon_1 = 1,\ \epsilon_2, \ldots, \epsilon_6 = \pm 1} \left(\prod_{r=1}^{6} \epsilon_r \right) \prod_{c=1}^{6} \left(\sum_{r=1}^{6} \epsilon_r x_{rc} \right).$$

Walsh cancellation leaves exactly the monomials that use each row once. This is a 32-term Chow decomposition. Theorem 1.2 follows.

7. Exact verification

From the repository checkout, run

```
python scripts/n6_exact_ordinary_chow_rank_32.py \
  --verify-json data/n6_exact_ordinary_chow_rank_32.json
python -m unittest tests.test_n6_exact_ordinary_chow_rank_32 -v
```

The first command derives the coordinate intersection maxima, all 45,696 coordinate cases behind (3.2), the actual normal-form derivative ranks and squarefree subspaces in (4.7)–(4.8), the low-span monomial floors, and the final cancellation. It also verifies the unequal formal and actual dimensions in Remark 4.2.

The Borel fixed-point reductions, prolongation identity, elementary directional arguments, symmetric image-span lemma, and filtration increment (6.3) are written proofs, not numerical inferences.

Remark 7.1 (Scope)**.** The theorem concerns ordinary Chow rank over characteristic zero. It does not determine border Chow rank or prove $\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(\mathrm{perm}_n) = 2^{n-1}$ for general n. Product rank and Chow rank are used synonymously in the relevant literature; see [2].

References

[1] D. G. Glynn, *The permanent of a square matrix*, European J. Combin. **31** (2010), no. 7, 1887–1891, doi:10.1016/j.ejc.2010.01.010.

[2] N. Ilten and Z. Teitler, *Product ranks of the* 3×3 *determinant and permanent*, Canad. Math. Bull. **59** (2016), no. 2, 311–319, doi:10.4153/CMB-2015-076-1.